MILITARY SPECIFICATION

VISCOSIMETER TUBE, UNIVERSAL

1. SCOPE

1.1 This specification covers a combined "Saybolt Universal" and "Furol" viscosimeter tube, suitable for determining the viscosity of both light and lubricating oils, in accordance with American Society for Testing and Materials (ASTM) Designation D88-56.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6630

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(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Designation D88-56 Standard Method of Test for Saybolt Viscosity.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS.

3.1 Material. Material shall be entirely of brass, of the highest quality throughout, and suitable in all respects for the purpose intended.

3.2 Style and design. The viscosimeter tube shall be designed and constructed so that it can be used to test both light and lubricating oils, and shall be in accordance with ASTM Designation D88-56, using either the "Universal" or "Furol" orifice, as applicable to the oil being tested. A spanner wrench that will insert and remove the orifices shall be furnished.

3.3 Components.

3.3.1 Tube. The tube and its upper and lower hubs shall be machined from solid brass bar stock. The lower hub shall be threaded internally with a thread that will accommodate interchangeable orifices. The bottom of this hub shall be finished with a smooth flange, an external thread, soft lead washers, and a companion nut of the same material as the tube. The bottom of the bore of the hub shall be machined to accommodate a cork stopper.

3.3.2 Orifices. The tube shall be furnished with hardened stainless-steel "Universal" tip and "Furol" tip orifices. The orifices shall be interchangeable on the tube.

3.4 Identification marking. The viscosimeter tube shall be permanently and legibly marked with the manufacturer's name or registered trademark and model number.

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3.5 Workmanship. Workmanship shall be first class throughout. The viscosimeter tubes shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

4.3 Examination. The viscosimeter tubes shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) viscosimeter, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each viscosimeter shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-636. As required, carton design shall include suitable interior fittings or sufficient cushioning to insure against shifting of, or damage to, the contents. Carton closure shall be suitable to prevent accidental opening.

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5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. The viscosimeters, packaged as specified in 5.1.2.1, shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Gross weight of the exterior container shall not exceed the weight limitation of the applicable box specification, and in no case shall exceed 200 pounds.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform to 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. The viscosimeters, packaged as specified in 5.1.2.1, shall be overpacked in an exterior container conforming to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table III; or PPP-B-636, type CF, class domestic. Closure of fiberboard boxes shall be in accordance with method I of appendix to box specification. Gross weight of the exterior container shall not exceed the weight limitation of the applicable box specification, and in no case shall exceed 200 pounds.

5.3 Marking.

5.3.1 Interior packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable level of packing (see 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

FOLD UNDER

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

POSTAGE FEES PAID
DEFENSE SUPPLY - AGENCY

OFFICIAL BUSINESS

DEFENSE SUPPLY AGENCY

DEFENSE MEDICAL SUPPLY CENTER

ATTN: DIRECTORATE OF TECHNICAL OPERATIONS (CODE TM5)

3RD AVENUE AND 29TH STREET

BROOKLYN, N.Y., 11232

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION MILITARY SPECIFICATION MIL-V-36402(DM) DATED 5 March 1965
VISCOSIMETER TUBE, UNIVERSAL

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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